

A fine-grained comparison of pragmatic language understanding in humans and language models

Jennifer Hu¹, Sammy Floyd^{1,2}, Olessia Jouravlev³, Evelina Fedorenko^{1,4}, Edward Gibson¹

¹Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology

²Department of Psychology, Sarah Lawrence College

³Department of Cognitive Science, Carleton University

⁴McGovern Institute for Brain Research, Massachusetts Institute of Technology

{jennhu, samfloyd, evelina9, egibson}@mit.edu

olessiajouravlev@cunet.carleton.ca

Abstract

Pragmatics and non-literal language understanding are essential to human communication, and present a long-standing challenge for artificial language models. We perform a fine-grained comparison of language models and humans on seven pragmatic phenomena, using zero-shot prompting on an expert-curated set of English materials. We ask whether models (1) select pragmatic interpretations of speaker utterances, (2) make similar error patterns as humans, and (3) use similar linguistic cues as humans to solve the tasks. We find that the largest models achieve high accuracy and match human error patterns: within incorrect responses, models favor literal interpretations over heuristic-based distractors. We also find preliminary evidence that models and humans are sensitive to similar linguistic cues. Our results suggest that pragmatic behaviors can emerge in models without explicitly constructed representations of mental states. However, models tend to struggle with phenomena relying on social expectation violations.

1 Introduction

Non-literal language understanding is an essential part of communication. For example, in everyday conversations, humans readily comprehend the non-literal meanings of metaphors (*My new coworker is a block of ice*), polite deceptions (*I love the gift*), indirect requests (*It's a bit cold in this room*), and irony (*Classy pajamas, dude!*). These phenomena fall under the broad label of **pragmatics**, which encompasses the aspects of meaning that go beyond the literal semantics of what is said (Horn, 1972; Grice, 1975; Yule, 1996; Levinson, 2000).

A long-standing challenge for NLP is to build models that capture human pragmatic behaviors.

The remarkable abilities of modern language models (LMs) have triggered a recent effort to investigate whether such models capture pragmatic meaning, both through philosophical arguments (Bisk et al., 2020; Bender and Koller, 2020; Potts, 2020; Michael, 2020) and empirical evaluations (Jeretic et al., 2020; Zheng et al., 2021; Tong et al., 2021; Liu et al., 2022; Ruis et al., 2022; Stowe et al., 2022). However, prior empirical studies have primarily evaluated LMs based on a binary distinction between pragmatic and non-pragmatic responses, providing limited insights into models' weaknesses. A model could fail to reach the target pragmatic interpretation in multiple ways – for example, by preferring a literal interpretation, or by preferring a non-literal interpretation that violates certain social norms. Understanding these error patterns can suggest specific directions for improving the models, and foreshadow where pragmatics might go awry in user-facing settings (e.g., Saygin and Cicekli, 2002; Dombi et al., 2022; Kreiss et al., 2022).

From a cognitive perspective, understanding the pragmatic abilities of LMs could also offer insights into humans. Human pragmatic language comprehension involves a variety of mechanisms, such as basic language processing, knowledge of cultural and social norms (Trosborg, 2010), and reasoning about speakers' mental states (Brennan et al., 2010; Enrici et al., 2019; Rubio-Fernandez, 2021). However, it remains an open question when language understanding relies on explicit mentalizing – which may be cognitively effortful – versus lower-cost heuristics (e.g., Butterfill and Apperly, 2013; Heyes, 2014). Because LMs lack explicit, symbolic representations of mental states, they can serve as a tool for investigating whether pragmatic competence can arise without full-blown mentalizing (e.g., belief updates in the Rational Speech Act framework; Frank and Goodman, 2012).

Code and data: <https://github.com/jennhu/lm-pragmatics>

In this paper, we perform a fine-grained comparison of humans and LMs on pragmatic language understanding tasks. Adopting the approach of targeted linguistic evaluation (e.g., Linzen et al., 2016; Futrell et al., 2019; Hu et al., 2020), our analysis serves two goals: assessing the pragmatic capabilities of modern LMs, and revealing whether pragmatic behaviors emerge without explicitly constructed mental representations. Our test materials are a set of English multiple-choice questions curated by expert researchers (Floyd et al., In prep), covering seven diverse pragmatic phenomena. We use zero-shot prompting to evaluate models with varying sizes and training objectives: GPT-2 (Radford et al., 2019), *Tk*-Instruct (Wang et al., 2022), Flan-T5 (Chung et al., 2022), and InstructGPT (Ouyang et al., 2022).

Through model analyses and human experiments, we investigate the following questions: (1) Do models recover the hypothesized pragmatic interpretation of speaker utterances? (2) When models do not select the target response, what errors do they make – and how do these error patterns compare to those of humans? (3) Do models and humans use similar cues to arrive at pragmatic interpretations? We find that Flan-T5 (XL) and OpenAI’s text-davinci-002 achieve high accuracy and mirror the distribution of responses selected by humans. When these models are incorrect, they tend to select the incorrect literal (or straightforward) answer instead of distractors based on low-level heuristics. We also find preliminary evidence that models and humans are sensitive to similar linguistic cues. Our results suggest that some pragmatic behaviors emerge in models without explicitly constructed representations of agents’ mental states. However, models perform poorly on humor, irony, and conversational maxims, suggesting a difficulty with social conventions and expectations.

2 Related work

Prior work has evaluated LMs’ ability to recognize non-literal interpretations of linguistic input, such as scalar implicature (Jeretic et al., 2020; Schuster et al., 2020; Li et al., 2021) or figurative language (Tong et al., 2021; Liu et al., 2022; Gu et al., 2022; Stowe et al., 2022). In a broad-scale evaluation, Zheng et al. (2021) test five types of implicatures arising from Grice’s (1975) conversational maxims, and evaluate their models after training on the task. In our work, we consider Gricean implica-

tures as one of seven phenomena, and we evaluate pre-trained LMs without fine-tuning on our tasks.

Similar to our work, Ruis et al. (2022) also use prompting to evaluate LMs on pragmatic interpretation tasks. They formulate implicature tests as sentences ending with “yes” or “no” (e.g., “Esther asked “Can you come to my party on Friday?” and Juan responded “I have to work”, which means no.”). A model is considered pragmatic if it assigns higher probability to the token that makes the sentence consistent with an implicature. In our work, models must select from multiple interpretations, enabling a detailed error analysis and comparison to humans. Ruis et al.’s materials also focus on indirect question answering as an implicature trigger, whereas we consider a broader range of pragmatic phenomena and utterance types.

Since pragmatic language understanding often draws upon knowledge of social relations, our tasks are conceptually related to benchmarks for evaluating social commonsense (e.g., Sap et al., 2019; Zadeh et al., 2019). These evaluations focus on the interpretation of actions and events, whereas we focus on the interpretation of speaker utterances. Another hypothesized component of pragmatics is Theory of Mind (ToM; Leslie et al., 2004; Apperly, 2011), or the ability to reason about others’ mental states. Benchmarks for evaluating ToM in models (e.g., Nematzadeh et al., 2018; Le et al., 2019; Sap et al., 2022) primarily focus on false-belief tasks (Baron-Cohen et al., 1985), which assess whether a model can represent the beliefs of another agent that are factually incorrect but consistent with that agent’s observations. LMs have been shown to succeed on some ToM tests (Kosinski, 2023) while failing on others (Sap et al., 2022; Ullman, 2023).

3 Evaluation materials

3.1 Overview of stimuli

Our evaluation materials are taken from Floyd et al.’s (In prep) experiments,¹ covering seven phenomena. Each item is a multiple choice question, with answer options representing different types of interpretation strategies. For most of the tasks, the question has three parts: a short story context (1-3 sentences), an utterance by one of the characters, and a question about what the character intended to convey.² Table 1 shows an example item for each

¹Materials can be found at https://osf.io/6abgk/?view_only=42d448e3d0b14ecf8b87908b3a618672.

²The exceptions are Humor and Coherence.

Task	Example query	Example answer options
Deceits	Henry is sitting at his desk and watching TV, and reluctantly switches off the TV with the remote control and picks up a textbook. Shortly after, his mother comes in the room and asks, "What have you been doing up here?" Henry responds: "Reading." Why has Henry responded in such a way?	1. Correct He does not want to get into trouble for not studying. 2. Literal He has been reading for some time. 3. DistractorLexicalOverlap He does not want to offend his mom by not reading the books that she gave him. 4. DistractorSocialConvention He wants his mom to believe that he has been watching TV.
Indirect speech	Nate is about to leave the house. His wife points at a full bag of garbage and asks: "Are you going out?" What might she be trying to convey?	1. Correct She wants Nate to take the garbage out. 2. Literal She wants to know Nate's plans. 3. DistractorAssociative She wants Nate to bring his friends over. 4. DistractorLexicalOverlap She wants Nate to spend more time with the family.
Irony	It is a holiday. Stefan and Kim are sitting in the backseat of the car. They are fighting all the time. Their father says: "Oh, it is so pleasant here." What did the father want to convey?	1. Correct He does not want to listen to his kids' arguments. 2. Literal He enjoys listening to his kids fighting. 3. DistractorAssociative AC gives them some needed cool. 4. DistractorNonSequitur He remembers about his wife's birthday.
Maxims	Leslie and Jane are chatting at a coffee shop. Leslie asks, "Who was that man that I saw you with last night?" Jane responds, "The latte is unbelievable here." Why has Jane responded like this?	1. Correct She does not want to discuss the topic that Leslie has raised. 2. Literal She thinks that it is the best latte in the town. 3. DistractorAssociative The man who Leslie saw makes unbelievable lattes. 4. DistractorNonLiteral A coffee break is not a good time to discuss men.
Metaphor	Andrew and Bob were discussing the investment company where Andrew works. Bob said: "The investors are squirrels collecting nuts." What does Bob mean?	1. Correct They buy stocks hoping for future profit. 2. Literal Squirrels were hired to work in the company. 3. DistractorNonLiteral The investors dress and eat well. 4. DistractorNonSequitur Bob is allergic to nuts. 5. DistractorPlausibleLiteral The investors enjoy picking nuts as much as squirrels do.
Humor	Martha walked into a pastry shop. After surveying all the pastries, she decided on a chocolate pie. "I'll take that one," Martha said to the attendant, "the whole thing." "Shall I cut it into four or eight pieces?" the attendant asked.	1. Correct Martha said, "Four pieces, please; I'm on a diet." 2. Literal Martha said: "Well, there are five people for dessert tonight, so eight pieces will be about right." 3. DistractorAssociative Martha said, "You make the most delicious sweet rolls in town." 4. DistractorFunny Then the attendant squirted whipped cream in Martha's face. 5. DistractorNeutral Martha said, "My leg is hurting so much."
Coherence	Mary's exam was about to start. Her palms were sweaty.	1. Correct Coherent 2. Incorrect Incoherent

Table 1: Sample item from each task in our evaluation. All items are originally curated by [Floyd et al. \(In prep\)](#).

task, with annotated answer options. Green labels indicate the target pragmatic interpretation.³ Blue labels indicate the literal interpretation. Red labels indicate incorrect non-literal interpretations, which are based on heuristics such as lexical similarity to the story, thus serving as distractor options.

Each task has 20-40 items, which were manually curated by expert researchers to cover a broad range of non-literal phenomena and elicit individual differences among humans. The stimuli were not specifically designed to require Theory of Mind

³We refer to these answer options as "Correct" throughout the paper. However, these answers are only "correct" in the sense of a normative evaluation. We acknowledge the wide variation in individual humans' abilities and tendencies to use non-literal language, which is not captured in our analyses. We thank an anonymous reviewer for highlighting this point.

reasoning (ToM). However, behavioral and neural evidence suggests that many of the tested phenomena rely on mentalizing processes. In Section 3.2, we briefly describe the role of ToM for each tested phenomenon, and how LMs' training corpora may provide linguistic cues to perform the tasks.

3.2 Tested phenomena

Deceits. Humans produce polite deceits ("white lies") in the service of social and personal relationships (e.g., [Camden et al., 1984](#)). Behavioral studies in young children suggest that understanding white lies requires interpretive ToM, or the ability to allow different minds to interpret the same information in different ways ([Hsu and Cheung, 2013](#)). Furthermore, the tendency to produce white

lies is linked to emotional understanding abilities, (Demedardi et al., 2021), and moral judgments about white lies are linked to second-order false-belief understanding (Vendetti et al., 2019).

The Deceits task presents a story with a white lie, and asks why the speaker has used this utterance. The underlying intentions behind polite deceits are rarely explicitly explained in text. As a result, it is unlikely that LMs learn a direct connection between the utterance and the speaker’s intention during training on static texts. However, instances of polite deceits in text corpora may be accompanied by descriptions of characters’ emotional states, which may indicate that speakers’ intentions differ from what is literally conveyed by their utterance. This highlights the importance of context in interpreting deceits, which we return to in Section 5.3.1.

Indirect speech. Humans often use language in a performative sense, such as indirectly requesting an action from other individuals (e.g., Austin, 1975; Searle, 1975). Indirect or polite speech comprehension has been captured by Rational Speech Act (RSA; Frank and Goodman, 2012) models, which characterize listeners as performing Bayesian inference about a speaker who chooses utterances based on a tradeoff between epistemic and social utility (Brown and Levinson, 1987; Yoon et al., 2016, 2020; Lumer and Buschmeier, 2022).

The IndirectSpeech task presents a story with an indirect request, and asks what the speaker intends to convey. Like deceits, it’s unlikely that indirect speech acts are explained in text data. However, indirect requests may be followed by descriptions of the completion of the implied request – for example, that someone closed a window after hearing the utterance “It’s cold in here”. Therefore, models may learn relationships between the utterances and desired outcomes through linguistic experience.

Irony. Humans use irony to convey the opposite of the semantic content of their utterance (Booth, 1974; Wilson and Sperber, 1992; Attardo, 2000; Wilson and Sperber, 2012). As such, irony has long been hypothesized to rely on social reasoning and perspective-taking (e.g., Happé, 1993; Andrés-Roqueta and Katsos, 2017). Indeed, human irony comprehension behaviors are captured by Bayesian reasoning models that take into account speakers’ affective goals (Kao and Goodman, 2014). In addition, neuroimaging studies suggest that irony interpretation relies on brain regions that are implicated

in classic ToM tasks (Spotorno et al., 2012).

The Irony task presents a story with an ironic statement, and asks what the character intends to convey. While ironic statements are also rarely explained in text, models could leverage accompanying cues such as descriptions of characters’ emotional states or a mismatch in sentiment.

Maxims of conversation. Grice (1975) proposes that communication follows a set of *maxims*: be truthful; be relevant; be clear, brief, and orderly; and say as much as needed, and no more. A prevailing theory is that listeners derive implicatures by expecting speakers to be cooperative (i.e., abide by the maxims) and reasoning about speakers’ beliefs and goals. Indeed, there is extensive evidence for RSA models capturing these implicatures, such as those arising from the maxims of *quantity* (Potts et al., 2016; Frank et al., 2018; Degen, 2023) and *manner* (Bergen et al., 2016; Franke and Jäger, 2016; Tessler and Franke, 2018).

The Maxims task presents a story with a character flouting one of Grice’s maxims, and asks why the character has responded in such a way. Based on linguistic input, it may be easy for LMs to recognize when a speaker is flouting a maxim – for example, if an utterance is particularly long, features an uncommon syntactic construction, or diverges semantically from the context. However, it is unclear whether LMs will be able to recover the speaker’s underlying intentions.

Metaphor. Metaphors (Lakoff and Johnson, 1980) are used to draw comparisons between entities in a non-literal sense. Metaphor understanding has been hypothesized to require mentalizing (Happé, 1993), and fine-grained metaphor comprehension behaviors are captured by RSA models where listeners and speakers reason about each others’ beliefs and goals (Kao et al., 2014).

The Metaphor task presents a story with a metaphor, and asks what the speaker intends to convey. For models, the challenges of metaphor comprehension include accessing world knowledge and forming abstract relationships between domains. However, it is possible that the relevant properties of the entities under comparison could emerge through linguistic experience.

Humor. Humor is one of the most distinctive aspects of human conversation, reflecting communicative goals with complex social function (Veatch, 1998; Martin and Ford, 2018). Neu-

roimaging studies suggest that joke understanding is supported by regions in the ToM brain network (Kline Struhl et al., 2018). Behavioral tests also reveal associations between ToM and humor abilities (Aykan and Nalçacı, 2018; Bischetti et al., 2019).

The Humor task presents a joke and asks which punchline makes the joke the funniest.⁴ Some theories argue that humor is triggered by linguistic incongruity effects (e.g., Deckers and Kizer, 1975), which might be straightforward for LMs to detect. Recent work has also shown that LMs can explain certain jokes (Chowdhery et al., 2022). However, some of Floyd et al.’s Humor items require complex world knowledge – for example, that slicing a pie into four versus eight pieces does not change the total amount of pie (see Table 1). As such, selecting the funniest punchline is a nontrivial task.

Coherence inferences. Humans also make pragmatic inferences beyond the sentence level – for example, by assuming that consecutive sentences form a logical or sequential relationship. Moss and Schunn (2015) and Jacoby and Fedorenko (2020) find that constructing these discourse relationships loads on regions of the ToM brain network, suggesting a role of ToM in coherence inferences.

The Coherence task presents a pair of sentences, and asks whether the pair forms a coherent story.⁵ We assume that LMs’ training data, which consists of naturalistic text, is primarily coherent. Therefore, we expect LMs to be able to distinguish between coherent and incoherent sentence pairs (for an in-depth study, see Beyer et al., 2021).

4 Experiments

4.1 Evaluation paradigm

Our evaluation paradigm uses *zero-shot prompting*. Prompting can easily be adapted to all of our seven tasks, allowing us to compare performance across tasks within a model. Prompting also allows us to present models with inputs that are nearly identical to the stimuli seen by humans in Floyd et al.’s experiments, whereas other methods would require converting the stimuli into task-specific formats. We choose zero-shot prompts in order to evaluate the knowledge that emerges through training, and not through *in-context adaptation* to the task.

⁴Unlike the other tasks, there is no speaker utterance.

⁵This task differs from the others in that there is no speaker utterance, and the answer options are identical across items (“Coherent” or “Incoherent”).

Model	# parameters	Training
GPT-2	117M	Autoregressive LM
Tk-Instruct (3B)	3B	Multitask
Tk-Instruct (11B)	11B	Multitask
Flan-T5 (base)	250M	Multitask
Flan-T5 (XL)	3B	Multitask
InstructGPT-3 (ada)	350M (est.)	Multitask, human feedback
text-davinci-002	Unknown	FeedME

Table 2: Models tested in our experiments.

Prompt structure. Each prompt consists of two parts: task instructions, and a query. The instructions are nearly identical to the instructions presented to humans in Floyd et al.’s experiments, prepended with the keyword “Task:”. The only other modification is that the original instructions had a final sentence of “Please answer as quickly as possible”, which we replaced with a sentence like “The answer options are 1, 2, 3, or 4”.⁶

For all tasks except Humor, the query consists of the scenario (prepended with keyword “Scenario:”) and question, and then the numbered answer options (prepended with “Options:”).⁷ The prompt concludes with the keyword “Answer:”. Full example prompts are given in Appendix A.

Evaluation. To evaluate a model on a given item, we feed the prompt to the model, and measure the model’s probability distribution over tokens conditioned on the prompt. We compare the probabilities of each answer token (e.g., “1”, “2”, “3”, or “4”) under this distribution. The model is considered correct on a given item if it assigns highest probability to the correct answer token, among all the possible answer tokens for that item.

We generated 5 versions of each item by randomizing the order of answer options. This was done to control for the base probabilities of the answer tokens. Since we do not analyze generated text, the model results themselves are deterministic.

4.2 Models

We test seven models across four model families, summarized in Table 2.⁸ As a baseline, we first test a base **GPT-2** model (117M parameters; Radford et al., 2019), which is trained on an autoregressive language modeling objective.

Second, we test a set of models which are based on T5 (Raffel et al., 2020) and instruction-finetuned

⁶The exact answer options changed according to the task.

⁷For the Humor task, the joke is prepended with “Joke:”, and the answer options are prepended with “Punchlines:”.

⁸All non-OpenAI models were accessed via Huggingface (Wolf et al., 2020) and run on a single NVIDIA A100 GPU.

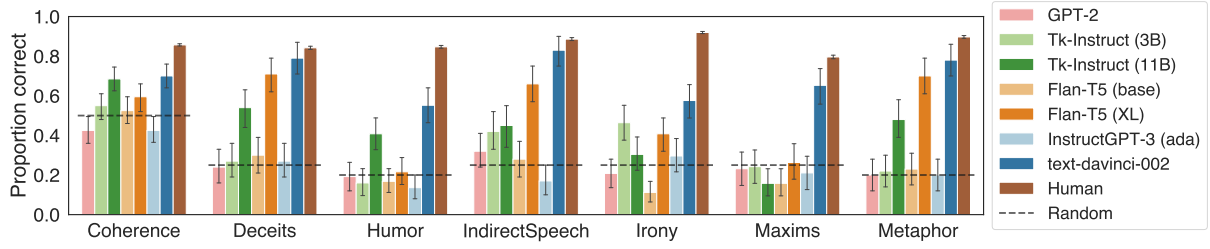


Figure 1: Accuracy for each task. Error bars denote 95% CI. Dashed line indicates task-specific random baseline.

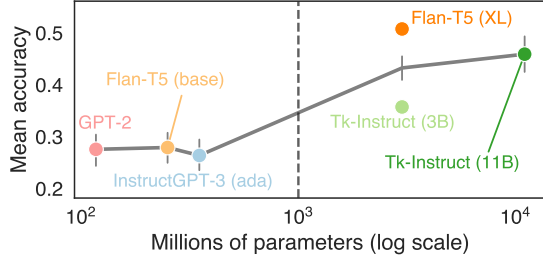


Figure 2: Mean accuracy vs. millions of parameters. Vertical dashed line indicates 1 billion parameters. text-davinci-002 was excluded from this analysis, as the number of parameters is unknown.

on a diverse collection of tasks (Wei et al., 2022). This set of models consists of two **Tk-Instruct** models (3B and 11B; Wang et al., 2022), which were fine-tuned on 1.6K tasks, and two **Flan-T5** models (base: 250M parameters; XL: 3B parameters; Chung et al., 2022), which were fine-tuned on 1.8K tasks. The fine-tuning tasks cover a wide range of categories, such as commonsense reasoning, translation, mathematics, and programming.

Finally, we test two **InstructGPT**-based models (Ouyang et al., 2022) via the OpenAI API: text-ada-001 (350M parameters), which we refer to as InstructGPT-3 (ada); and text-davinci-002, which comes from the GPT-3.5 family of models.^{9,10} These models are fine-tuned to follow instructions and align with human feedback.

We compare models to a baseline from 374 humans, collected by Floyd et al. (In prep). Their experiments presented multiple choice questions to humans in nearly identical format to our prompts.

5 Results

We now return to the three questions posed in the Introduction, in each of the following subsections.

⁹Parameter estimates come from <https://blog.eleuther.ai/gpt3-model-sizes/>. Although the size of text-davinci-002 is unknown, we assume that it is larger than InstructGPT-3 (ada).

¹⁰The OpenAI model results might not be reproducible, but

5.1 Do models choose the target pragmatic interpretation?

Figure 1 shows the proportion of trials where models and humans select the pragmatic answer. The smallest models (GPT-2, Flan-T5 (base), InstructGPT-3 (ada)) fail to perform above chance. The largest models (Tk-Instruct (11B), Flan-T5 (XL), text-davinci-002) perform above chance on all tasks (except Tk-Instruct (11B) on Maxims), and in some cases near human-level. Overall, models perform worst at the Humor, Irony, and Maxims tasks. Interestingly, these phenomena involve speakers violating listeners’ expectations in some way: producing a funny punchline to a mundane story (Humor), stating the direct opposite of the speaker’s belief (Irony), or disobeying one of the assumed rules of conversation (Maxims). It may be that models fail to represent certain social expectations that are maintained by human listeners.

Next, we investigated the relationship between model size and accuracy. Figure 2 shows the mean accuracy achieved by each model (averaged across tasks) vs. millions of parameters. The line and error bars denote the mean and 95% CIs, while points represent individual models. We find a coarse effect of model size: there is a stark jump in accuracy after 1B parameters (dashed line). However, model size does not fully explain variance in accuracy: all models with <1B parameters achieve similar accuracy, and Flan-T5 (XL) outperforms Tk-Instruct (3B), despite both having 3B parameters.

5.2 Do models and humans make similar types of errors?

Recall from Section 3 that each item has a set of answer options that correspond to different strategies (Table 1).¹¹ In addition to the target pragmatic answer (Correct), each item also has a plausible but unlikely literal answer (Literal), as well as distrac-

timestamps of API calls can be found in Appendix B.

¹¹The exception is Coherence, which is excluded here.

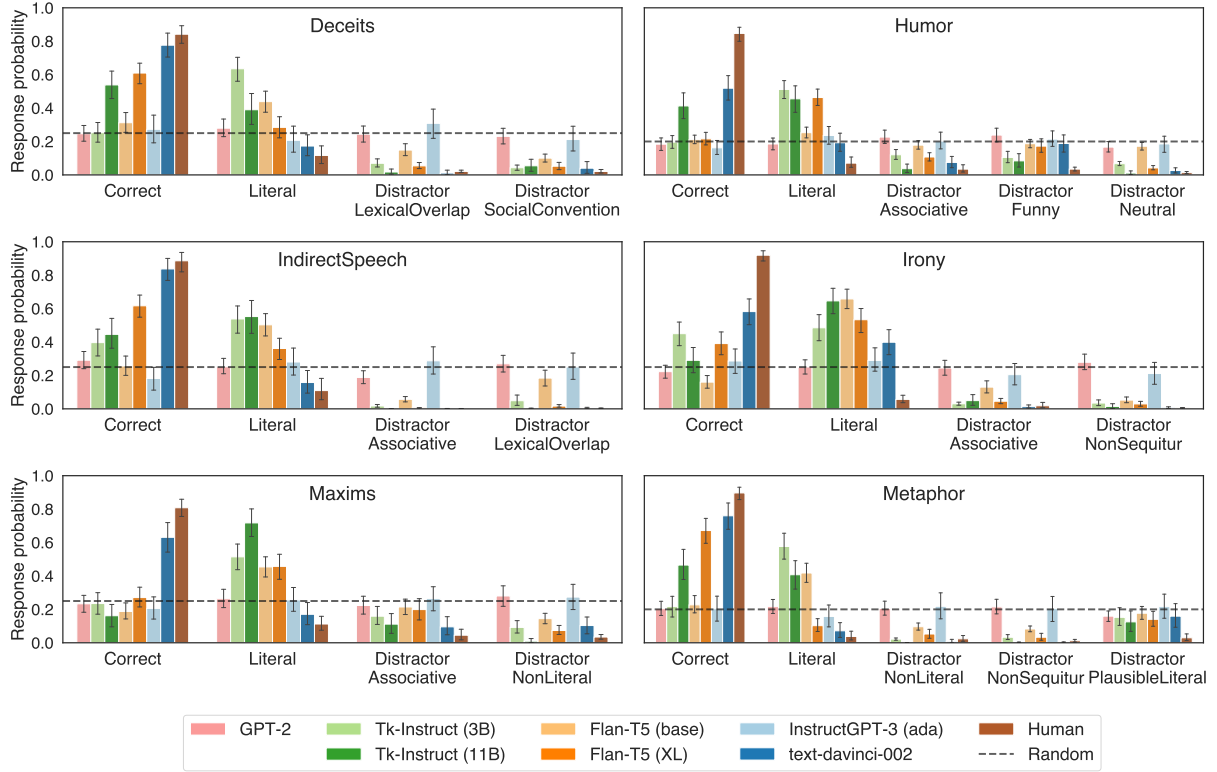


Figure 3: Response distributions across models and humans. Answer options for each task are shown on the x-axis. For models, y-axis denotes probability assigned to each answer option. For humans, y-axis denotes empirical frequency of each answer option being selected. Error bars denote 95% CI. Dashed line indicates random baseline.

tors based on lexical overlap or semantic associations (Distractor*). For each item, we computed the human empirical distribution over answer choices, and compared it to models’ probability assigned to the answer tokens (e.g., “1”, “2”, “3”, and “4”).

Figure 3 shows the answer distributions for each task. Across tasks, humans primarily select the Correct option, occasionally the Literal option, and rarely the distractors. We find a similar pattern for text-davinci-002, although the model is more likely to select the Literal option in general. The other large models (Tk-Instruct (11B), Flan-T5 (XL)) also generally assign highest probability to the Correct and Literal options, although the distribution looks less human-like. The next-largest models (Tk-Instruct (3B), Flan-T5 (base)) prefer the Literal option, and the remaining models (GPT-2, InstructGPT-3 (ada)) are at chance. These results show that larger models consistently identify the literal interpretation of an utterance, suggesting that their pragmatic failures are unlikely to be explained by a failure to represent basic semantic meaning (for our test materials).

However, even high-performing models occasionally do select the distractor answers, reveal-

ing interesting behaviors. For example, in the Metaphor task, text-davinci-002 and Flan-T5 (XL) prefer the DistractorPlausibleLiteral option – which is a figurative reading of the utterance – over the Literal option – which is completely non-figurative. Similarly, in the Humor task, text-davinci-002 is much more likely to select the DistractorFunny option over the other (non-humorous) distractors. This suggests a coarse sensitivity to humor, even if the model selects the human-preferred punchline only 55% of the time (see Figure 1). We take this analysis to illustrate the value of looking beyond binary pragmatic/non-pragmatic response distinctions, and using controlled distractor items to evaluate models’ abilities (e.g., McCoy et al., 2019).

5.3 Are models and humans sensitive to similar linguistic cues?

Having found qualitatively similar response patterns between humans and models, we now ask how models and humans arrive at pragmatic interpretations, and whether they use similar types of information. We begin with a broad evaluation of the extent to which models and humans rely on linguistic context (Section 5.3.1). We then take a

more granular approach and ask whether model and human performance is correlated at the item level – i.e., if models and humans exhibit similar sensitivity to the cues that make a non-literal interpretation more or less likely (Section 5.3.3).

5.3.1 The role of context

Many cues for enriched language understanding come from the context in which the speaker makes their utterance. However, some aspects of non-literal comprehension might arise given the utterance in isolation, while others are highly sensitive to specific contextual details (e.g., Levinson, 2000). Therefore, we expect that the degree to which humans rely on context to select non-literal interpretations will vary across the tested tasks.

To investigate this variation, we created a new set of stimuli by removing the context stories, leaving only the speaker utterance and final question (e.g., *Dan says, “The dog knocked it over.” Why has Dan responded in such a way?*).¹² We re-ran the human experiment on 30 participants, following the protocols of Floyd et al. (In prep)’s original experiment using the no-context modified materials.¹³ We also re-ran the three models that achieved highest accuracy on the original items: *Tk-Instruct* (11B), *Flan-T5* (XL), and *text-davinci-002*.

Figure 4 shows the mean accuracy difference on the original versus no-context versions of each item.¹⁴ We find that models and humans exhibit a similar qualitative pattern: removing the story leads to the largest degradation for Irony, followed by Deceits and Maxims. This aligns with our intuitions, because in these cases, speakers’ utterances can be interpreted either literally or as the complete opposite, based on the specific social situation (e.g., “It is so pleasant here”). In contrast, there are smaller degradations for IndirectSpeech and Metaphor. This suggests that some indirect requests are conventionalized (e.g., “I am getting cold”), although their interpretations may be facilitated by context (e.g., Gibbs, 1979). Similarly, this suggests that metaphor interpretation may draw more upon global knowledge than local context.

5.3.2 Scrambling

Next, we tested whether models rely on syntactic and discourse-level information from the con-

¹²This manipulation is not compatible with the Humor and Coherence tasks, so they are excluded from this analysis.

¹³Details can be found in Appendix C.1.

¹⁴See Figure 6 in Appendix C.2 for comparison of raw accuracy scores on the original and no-context items.

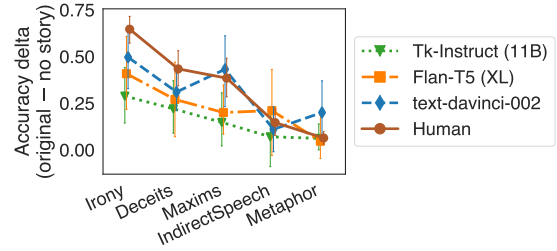


Figure 4: Mean by-item difference in accuracy once story context was removed.

text, or whether they can perform the tasks when ordering cues are removed. We constructed two scrambled versions of each item by randomizing the order of sentences and words. In both versions, the instructions, final question (e.g., *Why has Dan responded in such a way?*), and answer options were unmodified and remained in their original positions. Again, we only tested the best-performing models on these items.

We found that models maintain reasonable performance for most tasks, with the notable exception of Metaphor (Figure 7; Appendix D). This robustness to scrambling accords with prior evidence that models often rely on lexical information without human-like compositionality (e.g., Dasgupta et al., 2018; Nie et al., 2019; McCoy et al., 2019). We expect that scrambling, especially at the word-level, would likely disrupt human performance, but this remains an open empirical question. We leave an investigation of human performance to future work.

5.3.3 Item-level alignment

Up to this point, we analyzed differences across phenomena by averaging over items. However, there is also variance *within* each phenomenon in the types of cues that suggest how the utterances should be interpreted. For example, some items contain explicit descriptions of characters’ emotional states (e.g., “Sarah becomes angry”). If models and humans leverage these cues in similar ways, then we would expect to see correlations between model and human performance at the item level.

For each task and model, we compute the Pearson correlation between by-item mean accuracy achieved by humans and by-item mean probability that models assigned to the correct answer (Figure 5). In general, the larger models (*Tk-Instruct* (11B), *Flan-T5* (XL), *text-davinci-002*) are better aligned with humans, and the strongest correlations occur for IndirectSpeech, Irony, Maxims, and Metaphor. This suggests that for those tasks, mod-

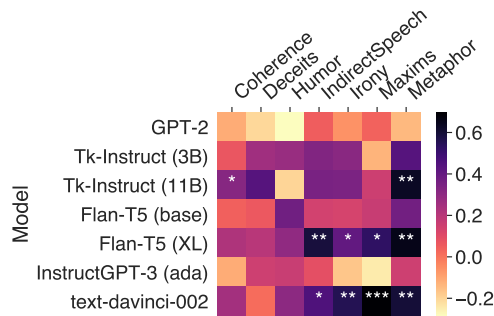


Figure 5: Pearson correlation coefficients between by-item human accuracy and model probability of the correct answer. Cells are marked with significance codes.

els and humans are similarly sensitive to cues that make a non-literal interpretation likely.

6 Discussion

We used an expert-curated set of materials (Floyd et al., In prep) to compare LMs and humans on seven pragmatic phenomena. We found that Flan-T5 (XL) and text-davinci-002 achieve high accuracy and match human error patterns: within incorrect responses, these models tend to select the literal interpretation of an utterance over heuristic-based distractors. We also found preliminary evidence that LMs and humans are sensitive to similar linguistic cues: model and human accuracy scores correlate at the item level for several tasks, and degrade in similar ways when context is removed.

Our results suggest that language models can consistently select the pragmatic interpretation of a speaker’s utterance – but how? The models tested in our experiments reflect a variety of learning processes through which pragmatic knowledge could emerge. GPT-2 is trained to learn the distribution of linguistic forms; the Tk-Instruct and Flan-T5 models are pre-trained on a denoising task and fine-tuned on thousands of instruction-based tasks; and the OpenAI models receive signal from human feedback. Our experiments are not designed to tease apart the contributions of these training procedures to models’ behaviors. Therefore, we do not intend to make strong claims about the mechanisms by which models learn pragmatics.

A shared feature of our tested models is the lack of explicitly constructed mental state representations. In this sense, our results are potentially compatible with two hypotheses. One possibility is that the models do not have an ability that can be considered an analog of Theory of Mind (ToM).

This view is supported by evidence that language models perform poorly on social commonsense and false-belief tasks (Sap et al., 2022), and are remarkably brittle to small perturbations of classic tests (Ullman, 2023). If models truly lack ToM, then their pragmatic behaviors might be explained by inferences based on low-level linguistic cues. Taken a step further, this finding could potentially suggest that certain human pragmatic behaviors arise through inferences based on language statistics, with no need for mental state representations.

A second possibility is that models do have a heuristic version of ToM, which is not explicitly engineered but instead emerges as a by-product of optimizing for other objectives (such as linguistic prediction). Since language contains many descriptions of agents’ beliefs, emotions, and desires, it may be beneficial – perhaps even necessary – to induce representations of these mental states in order to learn a generative model of linguistic forms. Indeed, Andreas (2022) argues that whereas language models have no explicit representation of communicative intents, they can infer approximate representations of the mental states of the agents that produce a given linguistic context. If this hypothesis is true, however, it would still remain unclear whether ToM is *necessary* to support the pragmatic behaviors tested in our evaluation materials.

Our experiments do not differentiate between these two hypotheses. However, fine-grained behavioral evaluations – such as those presented in this work – are important for revealing models’ capabilities and weaknesses, and offer a first step toward understanding how pragmatic behaviors can be supported. A promising direction for future work is to test models with a wider range of training objectives, or even new architectures, such as distinct language and social reasoning modules (see Mahowald et al., 2023). In addition, although there is evidence for the role of mentalizing in our tested pragmatic phenomena (see Section 3.1), one limitation of our stimuli is that they were not specifically designed to require ToM. New datasets that perform targeted manipulations of ToM alongside tests of language comprehension could help reveal how linguistic experience and ToM jointly support pragmatic behaviors.

Acknowledgments

We would like to thank the anonymous reviewers as well as Roger Levy, Christopher Potts, and

Josh Tenenbaum for their constructive feedback. We also thank Quinn Langford for help with coding details of the stimuli. This work was in part supported by a grant from the Simons Foundation to the Simons Center for the Social Brain at MIT. J.H. is supported by an NSF Graduate Research Fellowship (#1745302) and an NSF Doctoral Dissertation Research Improvement Grant (BCS-2116918). S.F. is funded by the NSF SPRF (#2105136). E.F. was additionally supported by NIH award R01-DC016950 and by research funds from the McGovern Institute for Brain Research and the Department of Brain and Cognitive Sciences.

Limitations

We note several methodological limitations with our experiments. **First**, since the evaluation materials were manually crafted, there is a rather small number of items (compared to the size of automatically generated NLP benchmarks). Small evaluation sets can introduce issues of statistical power (Card et al., 2020) and introduce bias based on lexical items. We feel this is not a major concern, because (1) our materials are validated by expert researchers; (2) models can be directly compared to humans in Floyd et al.’s experiments; and (3) in practice, there is enough signal to distinguish between the tested models.

Second, we only evaluate models on English-language materials, and some of the tasks were designed based on norms of communication and social interaction in Western cultures. As pragmatics can vary widely across language and cultures (Li, 2012; Rubio-Fernandez and Jara-Ettinger, 2020; Floyd, 2021; Brown et al., 2021; Dideriksen et al., 2022), an important direction for future work is to evaluate pragmatics beyond English (Ameka and Terkourafi, 2019; Blasi et al., 2022).

Third, aside from the OpenAI API models, we were only able to test models with $\leq 11\text{B}$ parameters due to limited computational resources. Models with parameter sizes between 11B and the size of text-davinci-002 could exhibit qualitatively different behaviors.

Finally, we emphasize that it is impossible to predict how models will respond to an arbitrary input. Therefore, we caution against extrapolating from our results and expecting that models will behave “pragmatically” in downstream applications. This is especially true for models behind the OpenAI API,

and text-davinci-002 in particular, for which very little is publicly known about the training protocol.

Ethics statement

Language technologies have the potential to cause harm at the individual and societal levels. Large language models (LLMs), which are typically trained on vast amounts of internet text, have been shown to perpetuate stereotypes based on gender, race, and sexual orientation. Applications using LLMs could reinforce systematic discrimination and amplify existing socioeconomic inequities. For example, LLMs could perpetuate social biases by assisting with hiring decisions or legal rulings.

The remarkable fluency of LLM-generated text also poses risks for the general public. LLMs have long been used to generate text that is difficult to distinguish from human-written text, raising concerns about detecting fake news and misinformation. Recently, LLMs have been used to synthesize knowledge – for example, by answering scientific questions (Taylor et al., 2022) or acting as search engines (Shah and Bender, 2022). Using LLMs as knowledge-providers could tremendously impact the nature of human collaboration and work, raising the need for model transparency and explainability.

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A Example prompts

This section contains example prompts for each task in our experiments. See Section 3 and Table 1 for details on the materials, and Section 4.1 for discussion of how prompts were constructed.

A.1 Deceits

Task: You will read short stories that describe two characters interacting with each other. Each story will be followed by a multiple-choice question. Read each story and choose the best answer to each question. Your task is to decide why the character in the story responds in a certain way. The answer options are 1, 2, 3, or 4.

Scenario: Henry is sitting at his desk and watching TV, and reluctantly switches off the TV with the remote control and picks up a textbook. Shortly after, his mother comes in the room and asks, "What have you been doing up here?" Henry responds: "Reading." Why has Henry responded in such a way?

Options:

- 1) He has been reading for some time.
- 2) He does not want to offend his mom by not reading the books that she gave him.
- 3) He does not want to get into trouble for not studying.
- 4) He wants his mom to believe that he has been watching TV.

Answer:

A.2 IndirectSpeech

Task: You will read short stories that describe everyday situations. Each story will be followed by a multiple-choice question. Read each story and choose the best answer. Your task is to decide what the character in the story is trying to convey. The answer options are 1, 2, 3, or 4.

Scenario: Nate is about to leave the house. His wife points at a full bag of garbage and asks: "Are you going out?" What might she be trying to convey?

Options:

- 1) She wants Nate to spend more time with the family.
- 2) She wants to know Nate's plans.
- 3) She wants Nate to take the garbage out.
- 4) She wants Nate to bring his friends over.

Answer:

A.3 Irony

Task: You will read short stories that describe everyday situations. Each story will be followed by a multiple-choice question. Read each story and choose the best answer. Your task is to decide what the character in the story is trying to convey. The answer options are 1, 2, 3, or 4.

Scenario: It is a holiday. Stefan and Kim are sitting in the backseat of the car. They are fighting all the time. Their father says: "Oh, it is so pleasant here." What did the father want to convey?

Options:

- 1) He enjoys listening to his kids fighting.
- 2) He remembers about his wife's birthday.
- 3) He does not want to listen to his kids' arguments.
- 4) AC gives them some needed cool.

Answer:

A.4 Maxims

Task: You will read short stories that describe everyday situations. Each story will be followed by a multiple-choice question. Read each story and choose the best answer. Your task is to decide why the character in the story responds in a certain way. The answer options are 1, 2, 3, or 4.

Scenario: Leslie and Jane are chatting at a coffee shop. Leslie asks, "Who was that man that I saw you with last night?" Jane responds, "The latte is unbelievable here." Why has Jane responded like this?

Options:

- 1) She does not want to discuss the topic that Leslie has raised.
- 2) The man who Leslie saw makes unbelievable lattes.
- 3) She thinks that it is the best latte in the town.
- 4) A coffee break is not a good time to discuss men.

Answer:

A.5 Metaphor

Task: You will read short stories that describe everyday situations. Each story will be followed by a multiple-choice question. Read each story and choose the best answer to each question. The answer options are 1, 2, 3, 4, or 5.

Scenario: Andrew and Bob were discussing the investment company where Andrew works. Bob said: "The investors are squirrels collecting nuts." What does Bob mean?

Options:

- 1) The investors dress and eat well.
- 2) Squirrels were hired to work in the company.
- 3) Bob is allergic to nuts.
- 4) They buy stocks hoping for future profit.
- 5) The investors enjoy picking nuts as much as squirrels do.

Answer:

A.6 Humor

Task: You will read jokes that are missing their punch lines. A punch line is a funny line that finishes the joke. Each joke will be followed by five possible endings. Please choose the ending that makes the joke funny. The answer options are 1, 2, 3, 4, or 5.

Joke: Martha walked into a pastry shop. After surveying all the pastries, she decided on a chocolate pie. "I'll take that one," Martha said to the attendant, "the whole thing." "Shall I cut it into four or eight pieces?" the attendant asked.

Punchlines:

- 1) Martha said, "My leg is hurting so much."
- 2) Martha said, "Four pieces, please; I'm on a diet."
- 3) Martha said: "Well, there are five people for dessert tonight, so eight pieces will be about right."
- 4) Then the attendant squirted whipped cream in Martha's face.
- 5) Martha said, "You make the most delicious sweet rolls in town."

Answer:

A.7 Coherence

Task: You will read pairs of sentences. Reach each pair and decide whether they form a coherent story. The answer options are 1 or 2.

Scenario: Cleo brushed against a table with a vase on it. She decided to study harder to catch up.

Options:

- 1) Incoherent
- 2) Coherent

Answer:

B Timestamps of OpenAI model queries

Table 3 shows timestamps of requests sent to the OpenAI API.

Model	Phenomenon	Timestamp
text-ada-001	Coherence	2022-10-11 12:28 -0400
text-ada-001	Deceits	2022-10-11 12:28 -0400
text-ada-001	IndirectSpeech	2022-10-11 12:28 -0400
text-ada-001	Irony	2022-10-11 12:28 -0400
text-ada-001	Humor	2022-10-11 12:28 -0400
text-ada-001	Maxims	2022-10-11 12:29 -0400
text-ada-001	Metaphor	2022-10-11 12:29 -0400
text-davinci-002	Coherence	2022-10-11 11:56 -0400
text-davinci-002	Deceits	2022-10-11 11:55 -0400
text-davinci-002	IndirectSpeech	2022-10-11 11:55 -0400
text-davinci-002	Irony	2022-10-11 11:54 -0400
text-davinci-002	Humor	2022-10-11 11:53 -0400
text-davinci-002	Maxims	2022-10-11 11:56 -0400
text-davinci-002	Metaphor	2022-10-11 11:57 -0400

Table 3: Timestamps of OpenAI API model queries.

C No-context analysis

C.1 Details of human experiments

Below, we discuss details of the no-context human experiments described in Section 5.3.1. This study was approved by the Institutional Review Board

at the home institution of the authors (protocol 2010000243).

Participants. We collected data from 30 participants using Amazon.com's Mechanical Turk. All participants were recruited from IP addresses in the US, Canada, and other English-speaking countries and passed a brief English proficiency task to participate. We pre-screened participants using a qualification task in which they were asked to perform 10 simple sentence completions, which were judged for basic levels of coherence and grammaticality. Participants were paid 7 USD for completing the study, which took around 20 minutes to complete. The resulting hourly rate was around 21 USD, which is well above federal minimum wage in the United States.

Procedure. Participants completed these tests during one individual testing session. After giving informed consent, which included assurance of anonymity, participants were shown instructions and a training trial, in which they were told they would be answering questions about a character in a short interaction. They then saw 105 trials (similar to those described in Appendix A), without the scenario context. For example:

Bob said: "The investors are squirrels collecting nuts." What does Bob mean?

- 1) The investors dress and eat well.
- 2) Squirrels were hired to work in the company.
- 3) Bob is allergic to nuts.
- 4) They buy stocks hoping for future profit.
- 5) The investors enjoy picking nuts as much as squirrels do.

Items were presented within blocks according to their phenomenon, as in Floyd et al.'s (In prep) original experiments. Blocks and items were presented in a random order.

C.2 Raw accuracy scores

Figure 6 shows accuracy scores achieved by humans and the three best-performing models on the original (shaded bars) and no-context (empty bars) versions of the test items.

D Sentence- and word-level scrambling

Figure 7 shows accuracy scores achieved by the three best-performing models on each task, across three scrambling conditions: none (original, unmodified items), sentence-level, and word-level. Example prompts are provided below.

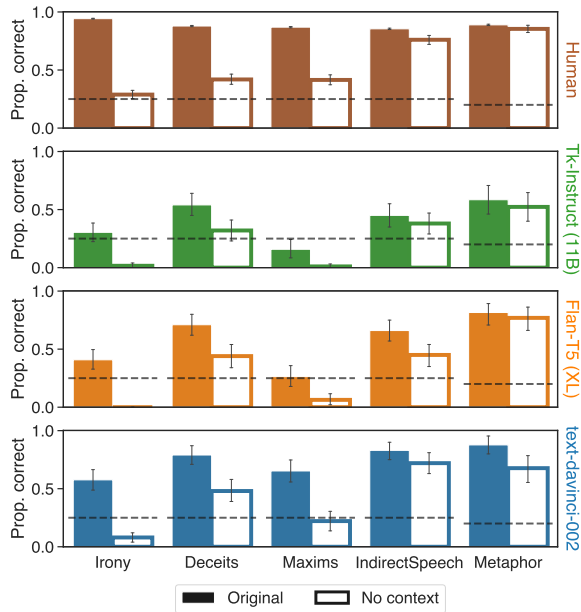


Figure 6: Proportion of items where humans and models select the correct pragmatic answer, on both original (shaded bars) and no-context (empty bars) versions.

D.1 Sentence-level scrambled prompt

Task: You will read short stories that describe two characters interacting with each other. Each story will be followed by a multiple-choice question. Read each story and choose the best answer to each question. Your task is to decide why the character in the story responds in a certain way. The answer options are 1, 2, 3, or 4.

Scenario: Dan says, "The dog knocked it over." The vase falls down on the floor and breaks. He brushes against his mother's vase. When Dan's mother comes home, she asks Dan: "What happened to my vase?" Dan is playing in the living room. Why has Dan responded in such a way?

Options:

- 1) Dan does not want his mom to be angry with him for breaking the vase.
- 2) Dan finds this vase ugly and wants to get rid of it.
- 3) Dan wants his mom to know that he knocked it over.
- 4) Dan thinks that the dog has knocked over the vase.

Answer:

4) Dan thinks that the dog has knocked over the vase.
Answer:

D.2 Word-level scrambled prompt

Task: You will read short stories that describe two characters interacting with each other. Each story will be followed by a multiple-choice question. Read each story and choose the best answer to each question. Your task is to decide why the character in the story responds in a certain way. The answer options are 1, 2, 3, or 4.

Scenario: to happened Dan "The against in it she comes "What living Dan the vase floor on down The Dan: He dog my brushes vase?" mother When falls breaks. vase. and playing room. his asks knocked says, home, over." the mother's is Dan's Why has Dan responded in such a way?

Options:

- 1) Dan does not want his mom to be angry with him for breaking the vase.
- 2) Dan finds this vase ugly and wants to get rid of it.
- 3) Dan wants his mom to know that he knocked it over.

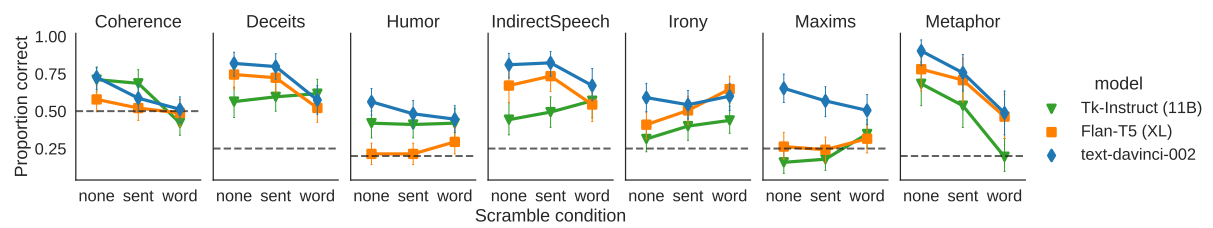


Figure 7: Model performance across scrambling conditions (none = original, unmodified items). Error bars denote 95% CI. Dashed line indicates random baseline.